

Replication of Study X by Sample & Sample (20xx, Psychological Science)

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Introduction

Proposal: Justification Zettersten & Saffran (2020) uses cross situational learning tasks to understand how learners gather new information for word learning. It applies an active learning and sampling lens to word learning. Experiment 1 aims to study whether adults engage in information seeking behavior that helps in disambiguating the reference, which is the central question they later ask in Experiment 2 with a developmental population. I am to replicate Experiment 1 (including the results in supplementary analyses s1). Ultimately I am also

interested in a yoked version of this task to understand whether active learners outperformed their yoked counterparts on the same items.

Methods Stimuli is available on OSF (alien images, novel words). The experiment was also developed in jsPsych (older version). In the task, there are three phases: training phase (where two referents and two labels are shown in random order, occurring 6 times, for a total of 24 trials), sampling phase (objects appear in 8 randomized locations), recall test without feedback

Challenges The experiment was designed using some custom jsPsych plugins, which is where I believe I might run into some challenges.

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or “none”.

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.